

Dividing a natural number by a proper fraction

How many halves are in three? — the division that makes the answer bigger.

LESSON 24–26

3 × 40 MIN

MATEMATIKA 6, §6

S7 7.4

LESSON CLOCK

25'

25'

30'

30'

10

How many fit?	25 min
The reciprocal	25 min
Why the answer grows	30 min
Problems	30 min
Homework	10 min

LEARNING OBJECTIVES

- Interpret dividing by a fraction as “how many of these fit”.
- Divide a whole number by a fraction using the reciprocal.
- Explain why the answer is larger than the number divided.
- Solve measuring and packing problems.

TEXTBOOK

National	pp. 65–72
Cambridge	Stage 7, pp. 71–75
Workbook	Exercise 7.4

01

Explanation

How many fit?

$3 \div \frac{1}{2}$ asks: how many halves are there in 3? Six — and a picture settles it before any rule is stated.

$$3 \div \frac{1}{2} = 6$$

DIVISION	THE QUESTION IT ASKS	ANSWER
$3 \div \frac{1}{2}$	how many halves in 3	6
$4 \div \frac{1}{4}$	how many quarters in 4	16
$6 \div \frac{2}{3}$	how many two-thirds in 6	9
$5 \div \frac{1}{3}$	how many thirds in 5	15

DIVISION HAS TWO MEANINGS, AND THIS IS THE SECOND

$12 \div 3$ can mean “share 12 among 3” or “how many 3s are in 12”. With a fraction as the divisor, only the second question makes sense.

The reciprocal

$$n \div \frac{a}{b} = n \cdot \frac{b}{a}$$

Turn the divisor upside down and multiply. The **reciprocal** of $\frac{a}{b}$ is $\frac{b}{a}$, and the two multiply to 1.

FRACTION	RECIPROCAL	PRODUCT
$\frac{1}{2}$	2	1
$\frac{2}{3}$	$\frac{3}{2}$	1
5	$\frac{1}{5}$	1
$\frac{7}{4}$	$\frac{4}{7}$	1

DIVISION	AS MULTIPLICATION	ANSWER
$6 \div \frac{2}{3}$	$6 \cdot \frac{3}{2}$	9
$4 \div \frac{4}{5}$	$4 \cdot \frac{5}{4}$	5
$10 \div \frac{5}{6}$	$10 \cdot \frac{6}{5}$	12

TURN OVER THE DIVISOR, NEVER THE DIVIDEND

$6 \div \frac{2}{3}$ is $6 \cdot \frac{3}{2} = 9$. Turning over the 6 instead gives $\frac{1}{9}$ — the reciprocal of the right answer, and a mistake the estimate below catches instantly.

Why the answer grows

Dividing by a number smaller than 1 always gives a larger answer, because the pieces being counted are smaller than the whole.

DIVISOR	COMPARED WITH 1	THE ANSWER IS	EXAMPLE
$\frac{1}{2}$	smaller	larger than 6	$6 \div \frac{1}{2} = 12$
1	equal	unchanged	$6 \div 1 = 6$
$\frac{3}{2}$	larger	smaller than 6	$6 \div \frac{3}{2} = 4$

ESTIMATE BEFORE YOU CALCULATE

Ask first: will the answer be bigger or smaller than the number I started with? One second of thought rules out half the possible mistakes.

Problems

PROBLEM	WORKING	ANSWER
6 m of cloth cut into pieces of $\frac{2}{3}$ m	$6 \div \frac{2}{3}$	9
4 litres poured into $\frac{1}{4}$ -litre cups	$4 \div \frac{1}{4}$	16
10 kg of rice in $\frac{5}{6}$ -kg bags	$10 \div \frac{5}{6}$	12
a wall 9 m long, tiles $\frac{3}{4}$ m wide	$9 \div \frac{3}{4}$	12

“HOW MANY ... FIT” IS THE PHRASE TO LOOK FOR

Cutting, pouring, packing and tiling are all the same question. Once the phrase is spotted, the division writes itself.

02

Worked examples

EXAMPLE 1

Compute $6 \div \frac{2}{3}$.

1

The divisor is less than 1, so expect an answer over 6.

2

$6 \cdot \frac{3}{2}$

Turn over the divisor.

3

$= \frac{18}{2} = 9$

ANSWER

9

EXAMPLE 2 How many $\frac{3}{4}$ m tiles fit along a wall 9 m long?

① The question is “how many fit”, so divide.

② $9 \div \frac{3}{4} = 9 \cdot \frac{4}{3}$

③ $= 12$

ANSWER

12 tiles

EXAMPLE 3 Compute $10 \div \frac{5}{6}$.

① $10 \cdot \frac{6}{5}$

② Cancel: $10 \div 5 = 2$.

③ $2 \cdot 6 = 12$

ANSWER

12

03 Interactive model

Lay a metre rule on the desk and count how many 25-cm cards fit along it; the class answers “four” without hesitation and has just divided by a quarter.

Dividing by a fraction

$$6 \div \frac{2}{3}$$

- ① The divisor is under
1, so the answer beats
6.

Estimate
first.

$$\textcircled{2} \quad 6 \cdot \frac{3}{2}$$

Turn
over
the
divisor.

$$\textcircled{3} \quad 9$$

How
many
two-
thirds
fit in
six.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Reciprocal	Teskari son	Обратное число
How many fit	Nechta joylashadi	Сколько помещается
To turn upside down	Ag'darib qo'yish	Перевернуть
Divisor	Bo'luvchi	Делитель
Quotient	Bo'linma	Частное
Measuring problem	O'lchash masalasi	Задача на измерение
Whole number	Butun son	Целое число
Estimate	Baholash	Оценка

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$3 \div \underline{1} \ 2$ equals:

$$\frac{3}{2}$$

6

$$\frac{1}{6}$$

1.5

The reciprocal of $\underline{2} \ 3$ is:

$$\frac{3}{2}$$

$$-\frac{2}{3}$$

$$\frac{2}{3}$$

3

Dividing by a number under 1 gives an answer:

smaller

larger

the same

negative

$6 \div \underline{2} \ 3$ equals:

4

9

$\frac{4}{9}$

12

$4 \div \underline{4} \ 5$ equals:

5

$\frac{16}{5}$

$\frac{5}{4}$

3.2

Which fraction is turned over?

the first

the divisor

both

neither

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $3 \div \frac{1}{2}$

ANSWER 6

2. $4 \div \frac{1}{4}$

ANSWER 16

3. $5 \div \frac{1}{3}$

ANSWER 15

4. $6 \div \frac{2}{3}$

ANSWER 9

5. $4 \div \frac{4}{5}$

ANSWER 5

6. The reciprocal of $\frac{3}{7}$

ANSWER $\frac{7}{3}$

7. The reciprocal of 5

ANSWER $\frac{1}{5}$

Medium · the standard question

7 PROBLEMS

1. $10 \div \frac{5}{6}$

ANSWER 12

2. $9 \div \frac{3}{4}$

ANSWER 12

3. $8 \div \frac{2}{5}$

ANSWER 20

4. 6 m of cloth in $\frac{2}{3}$ m pieces

ANSWER 9

5. 4 litres in $\frac{1}{4}$ -litre cups

ANSWER 16

6. 10 kg in $\frac{5}{6}$ -kg bags

ANSWER 12

7. $6 \div \frac{3}{2}$

ANSWER 4

Hard · stretch and reason

7 PROBLEMS

1. $7 \div \frac{7}{9}$

ANSWER 9

2. $12 \div \frac{8}{9}$

ANSWER $13\frac{1}{2}$

3. $1 \div \frac{4}{7}$

ANSWER $\frac{7}{4}$

4. Which is larger, $8 \div \frac{1}{2}$ or $8 \cdot \frac{1}{2}$?

ANSWER $8 \div \frac{1}{2} = 16$

5. A wall 15 m long, tiles $\frac{5}{8}$ m wide

ANSWER 24 tiles

6. Why does dividing by $\frac{1}{2}$ double a number?

ANSWER Two halves make each whole

7. $20 \div \frac{4}{5} \div 5$

ANSWER 5

07 Homework

Homework — 5 tasks

Estimate first: will the answer be larger or smaller than the number you started with?

1. Compute $8 \div \frac{1}{2}$ and $9 \div \frac{1}{3}$.
2. Compute $12 \div \frac{3}{4}$ and $15 \div \frac{5}{6}$.
3. How many $\frac{2}{5}$ -litre bottles can be filled from 6 litres?
4. A ribbon 8 m long is cut into $\frac{4}{5}$ m pieces. How many pieces?
5. Explain why $6 \div \frac{2}{3}$ is larger than 6.